

ENGR 225 Homework 2 Questions

Assoc. Prof. Clayton Byers

Assigned: 18 Sept 2026

Due: 25 Sept 2026 (11:59pm on Moodle)

Important notes before you begin:

- Don't forget to follow the homework formatting guidelines!
- Also make sure you include the cover sheet with each submission.
- I am human and will make mistakes & typos from time to time. If you notice any typos or if you are positive a solution I have given in these problem sets is wrong, please let me know!

Problem 1

The towline exerts a force of $P = 6\text{ kN}$ at the end of the 8 m long crane boom. If $\theta = 30^\circ$, determine the placement x of the hook at B so that this force creates a maximum moment about the point O . What is the value of the moment? (**Ans:** $x = 9.81\text{ m}$, $M = 48.0\text{ kN} \cdot \text{m}$)

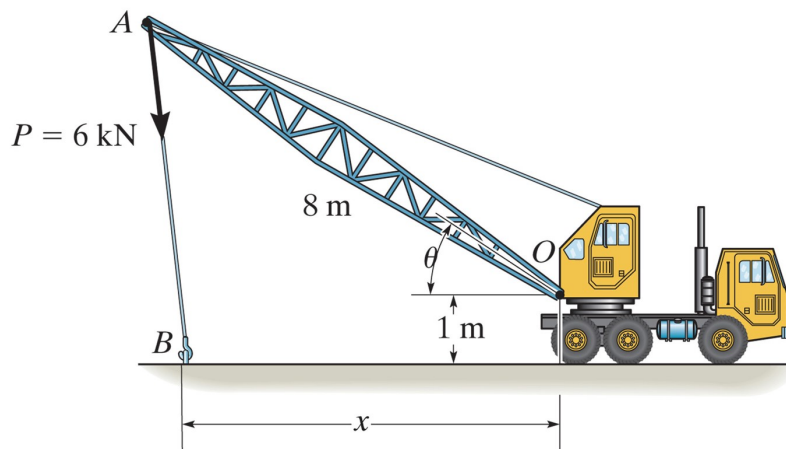


Figure for problem 1.

Problem 2

The crane is used to hoist a $2 Mg$ load upwards at a constant velocity. The $1.5 Mg$ jib BD and $0.5 Mg$ jib BC have centers of mass at G_1 and G_2 , respectively. Determine the required mass of the counterweight C so that the resultant moment produced by the load and the weight of the tower crane jibs about point A is zero. Assume the center of mass of the counterweight is at G_3 . (**Ans:** $m = 4.97 Mg$)

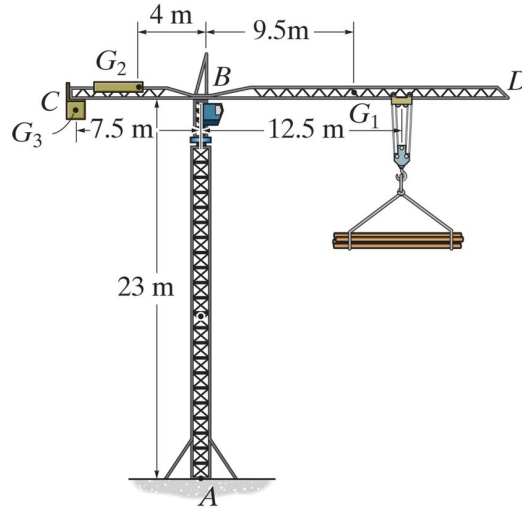


Figure for problem 2.

Problem 3

The force $\vec{F} = 6\hat{i} + 8\hat{j} + 10\hat{k} \text{ N}$ creates a moment about the point O of $M_O = -14\hat{i} + 8\hat{j} + 2\hat{k} \text{ N} \cdot \text{m}$. If the force passes through a point with an x coordinate of 1 m , determine the y and z coordinates of the point. Also, determine the perpendicular distance d from the point O to the line of action of the force $\vec{F}$. (**Ans:** $\vec{r} = 1\hat{i} + 1\hat{j} + 3\hat{k} \text{ m}$ and $d = 1.15 \text{ m}$)

Note: The figure has arbitrarily drawn vectors, they don't represent the actual values.

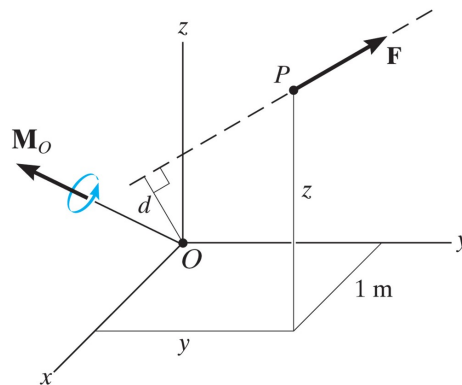


Figure for problem 3.

Problem 4

Determine the magnitude of the moment of the force $\vec{F} = 50\hat{i} - 20\hat{j} - 80\hat{k}$ N about the base line CA of the tripod. (**Ans:** $M_{CA} = 226 \text{ N} \cdot \text{m}$)

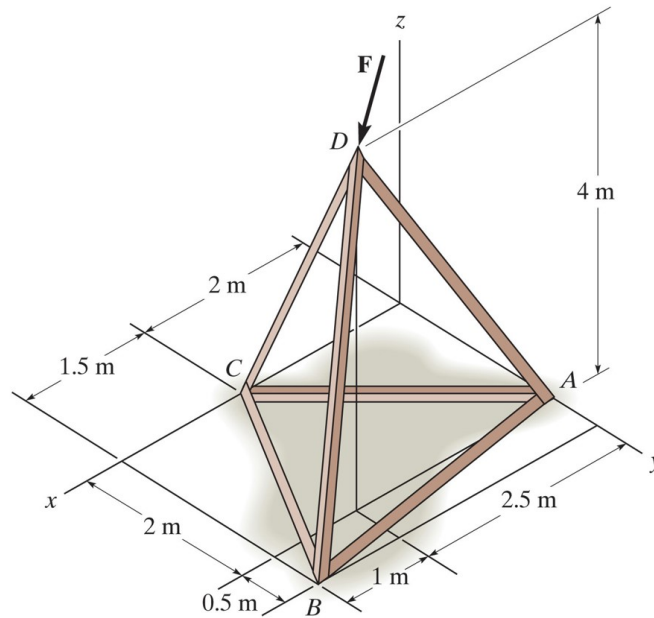


Figure for problem 4.

Problem 5

Determine the magnitude of the couple force $\vec{F}$ so that the resultant couple moment on the crank is zero. (**Ans:** $F = 194 \text{ lb}$)

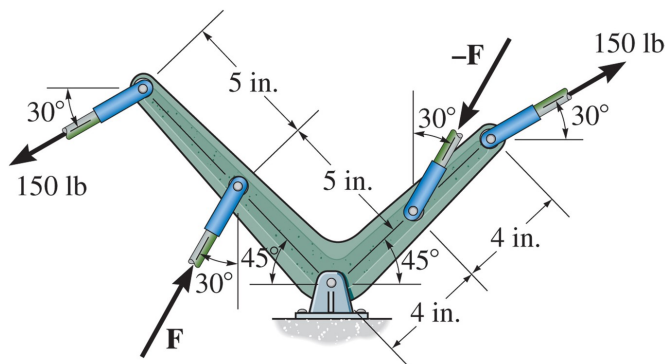


Figure for problem 5.

Problem 6

Replace the force system acting on the frame by an equivalent resultant force and couple moment acting at point A . (**Ans:** $F_R = 938\text{ N}$, $\theta = 35.9^\circ$ below x axis, and $M_A = 680\text{ N} \cdot \text{m}$ (CCW))

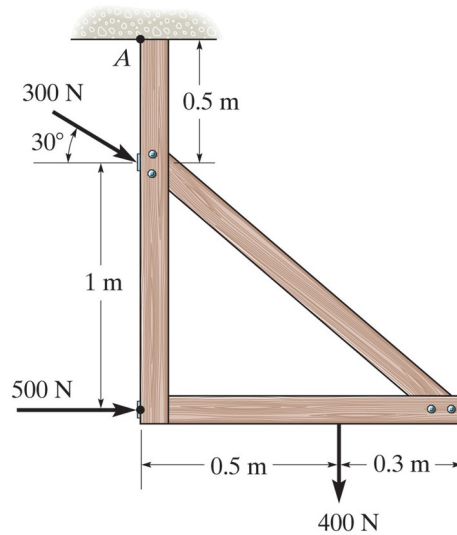


Figure for problem 6.

Problem 7

Replace the force system acting on the post by a resultant force, and specify where its line of action intersects the post AB measured from point B . (**Ans:** $\vec{F}_R = -533\hat{i} + 100\hat{j}\text{ N}$, $r_b = 2.17\text{ m}$ below B)

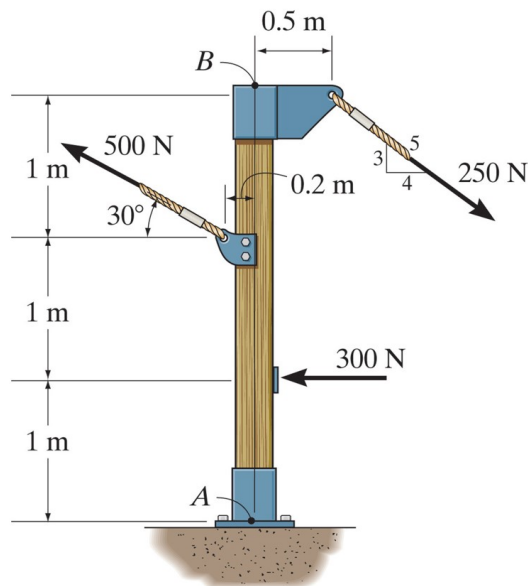


Figure for problem 7.

Problem 8

If $F_A = 40\text{ kN}$ and $F_B = 35\text{ kN}$, determine the magnitude of the resultant force and specify the location of its point of application (x, y) on the slab. (**Ans:** $\vec{F}_R = -215\hat{k}\text{ kN}$, $(x, y) = (3.54, 3.68)\text{ m}$)

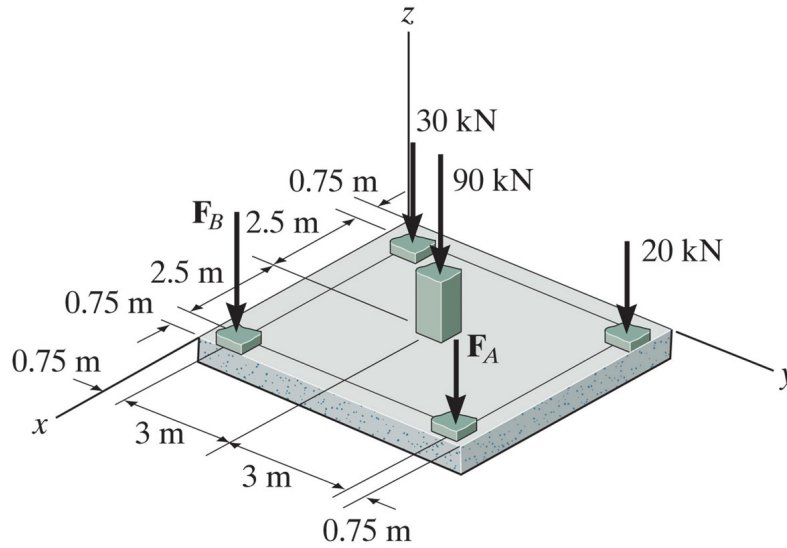


Figure for problem 8.